

Reviewer Comment Progress Summary

Current evidence reconciliation - 2 September 2026

Assessment basis. This update combines the current manuscript and response letter with retained results through 2 September 2026: merged Issues 33-36, the Issue 30 gate worktree, and exploratory chemical-potential-film work. A branch/worktree result is counted only at its stated claim boundary. Scores are internal assessments, not reviewer-assigned grades. Scale: 5 = complete, 4 = strong, 3 = partial, 2 = weak, 1 = absent.

Category Assessment

Category	Comments	Status	Before	Current	Current evidence and remaining work
Thermodynamic theory and scope	R1.1-R1.4	Addressed	1.5, F	4.8, A	Scope separates reaction equilibrium, transport, and molecular fugacity; the workflow and model context are explicit.
Parameters, provenance, and sensitivity	R1.5-R1.8; R2.3	Partial	1.4, F	3.0, C	The hash-consistent candidate bundle and 131-row regression are retained, but basis admission, independent thermodynamic validation, uncertainty/covariance, complete provenance, and pressure/VLE evidence remain open. No predictive parameter set is authorized for column transfer.
Transport-property uncertainty	R1.9	Partial	2.0, D	3.5, C+	Issues 35/42 catalogue source correlations, units, domains, and uncertainty statements. A provisional PSD Onsager closure now passes conservation/current/entropy checks, but species-resolved ionic diffusivities, source-complete cross-friction data, and propagated transport uncertainty remain absent; no physical film flux is compared.
Numerical convergence and cost	R1.10-R1.11	Partial	2.0, D	3.0, C	Fixed-chemistry rows and diagnostic BVP/film calculations run, but activity-coupled chemistry is not a production lane. Iteration/refinement counts, memory, and same-hardware profiling remain incomplete.
Other amines and transferability	R1.12; R2.6	Partial	1.0, F	3.7, B	Solvent-specific requirements are described. No second-solvent demonstration or held-out calibration workflow has been shown.
Limitations and model context	R1.13-R1.14; R2.2	Addressed	2.0, D	4.8, A	Claim boundaries, literature gap, and comparisons with eNRTL, Kent-Eisenberg, CPA, and ePC-SAFT are explicit.
Formatting and references	R2.1	Addressed	2.0, D	5.0, A	Numeric references are in place and citations begin with [1].
Validation breadth	R2.4	Partial	2.5, D	3.4, C+	Eighteen fixed-chemistry rows pass across K18, K19, and 1C-7C; K20 is rejected. Issue 40 evaluates one packet state but admits zero scientific rows; independent cross-facility validation remains absent.
Operating study and optimization	R2.5	Open	1.0, F	1.0, F	Still needed: a controlled operating envelope, constraints, objective, sensitivity study, and reproducible optimization.
Fully predictive reactive ePC-SAFT	Cross-cutting endpoint	Open	1.0, F	1.0, F	The bundle is loadable and packet-replayable, but Issue 40 has 0/5 scientifically admitted rows; Issues 41/42 and release Issue 36 remain supported-negative or blocked. No validated predictive reactive chemistry or column transfer is established.
Physical reactive film and column transfer	Issues 30, 36, 42	Open	1.0, F	1.0, F	Issue 30 records seven predeclared classes as not attempted because dependencies are blocked. No physical film flux, model disagreement, or packed-column adoption is demonstrated.

Overall Assessment

Assessment	Before	Current	Interpretation
Overall response to all requested reviewer items	1.8 / 5, D	3.1 / 5, C	Theory, scope, limitations, fixed-chemistry reproducibility, source records, and formatting improved. The current evidence does not support predictive reactive chemistry, physical film validation, independent external validation, or optimization.
Current re-scoped fixed-chemistry benchmark	1.0 / 5, F	4.3 / 5, B+/A-	The benchmark has a clear claim boundary and stronger reproducibility evidence: 18 accepted rows across K18, K19, and 1C-7C for both closures, with K20 retained as rejected.
Fully predictive reactive ePC-SAFT endpoint	1.0 / 5, F	1.0 / 5, F	A loadable bundle, stable packet replay, and converged prototypes are feasibility/readiness evidence only. The endpoint still lacks admitted basis, complete kinetics and transport, pressure-domain replay, independent validation, and column transfer.
Physical reactive-film / column-adoption gate	1.0 / 5, F	1.0 / 5, F	Issue 36 has 0/5 admitted film-input states and a blocked release; Issue 30 therefore launches no physical campaign. Exploratory source and aggregate film evidence does not establish a film model or capture result.

Verified Updates and Claim Boundaries

- **Genuine progress:** the fixed-chemistry manuscript is reconciled, with 18 accepted rows and K20 visibly rejected; Issues 33-36 preserve source-faithful mapping, kinetics, transport, and release records; and the immutable candidate bundle is internally hash-consistent.
- **Issue 40:** one nine-species packet state is numerically replayable, but all five declared rows remain basis-unresolved and the scientific-admission count is zero.
- **Issues 41/42 and 36:** source-rate and transport evidence is retained, but detailed balance, reaction timescales, physical transport, uncertainty propagation, and film-input release are not attempted; Issue 36 creates zero release files.
- **Issue 30:** all seven predeclared validation classes are not attempted because the dependency gate is blocked. This is a supported-negative no-run decision, not evidence of model agreement or disagreement.
- **Reactive formulation diagnostic:** the vendored bundle gives a fresh homogeneous equilibrium with reaction-affinity residual 1.7×10^{-13} ; its exact tangent is symmetric and positive-curvature, and the provisional Onsager closure enforces zero total flux/current with positive entropy production. A reaction-to-diffusion time-scale ratio of about 7.6×10^3 identifies a stiff fast-reaction manifold. These are internal consistency checks only; the assumed ion diffusivities, source-basis rate prefactors, row-level flux validation, and packed-column capture evidence remain absent.
- **Exploratory film branch:** Putta aggregate AARD values, Dugas dimensional film rows, and tracer/self-diffusion anchors improve source awareness only. They are not promoted to predictive or validation claims.

Bottom line. The revised record is materially stronger as a bounded fixed-chemistry benchmark and as a map of what is missing. It is not materially closer to an accepted fully predictive reactive ePC-SAFT absorber: that endpoint remains open at 1.0/5 until the scientific admission and downstream film/column gates pass.